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MULTI-PURPOSE SEMI-SUBMERSIBLES

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F A C S I M I L E

FROM: CRAIG LANG
TO: RIOMARINE
ATTN OF: RONALD CARELLI
FAX NO: 010 55 21 233 5331
NO. OF PAGES INC. THIS ONE: 2
DATE: 28TH NOVEMBER 1989

Dear Ronald

RE: FAX 5942/2 OF 28TH NOVEMBER 1989

The system proposed was used by Cammel-Laird to build the Sovereign Explorer. A special crane was used at a cost of £10,000 per day to assemble the column and deck boxes. The pontoons were built on ~~ships~~ launched and placed in parallel drydocks. The dock wall between the docks was excavated so that the cross-braces could be installed. Sponson tanks on the inside of the pontoons were added, buoyancy tanks from the pontoon decks to 6m above pontoon deck were added and finally more buoyancy tanks were added. These were added on the forward and aft faces of all columns. Total number of additional tanks, 18 in all. Working this way is extremely awkward and expensive. Access for outfitting by personnel is difficult.

Cammel Laird's bid £80 million, the final cost was £160 million. In their support, it was the first time they had built a semi and the Sovereign Explorer is unusually complicated since it has a combined chain and cable mooring system, designed for 50 - 600m (recently used for 1000m and being modified for 1700m); a drilling rig designed to drill 8000m into the seabed in 600m (the derrick had to be increased from 500te to 700te, with a big impact on the main deck sub-structure, and on and on! It almost came to a court case between Cammel Laird and GVA (I believe it was settled out of court).

The Balmoral FPV was also built in two drydocks. Again, outfitting problems were extremely difficult. The vessel cost rose from £125,000,000 to £160,000,000. I will send you details of all the change orders if you need them. The £160,000,000 does not include the cost of the process equipment, drilling, diving which were free issued by Bechtel to GVA (and paid for by Sun Oil, of course)!

continued.....

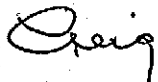
A large part of the problem came from the hazardous areas (and even the non-hazardous areas), created when process equipment was placed in the tween decks to keep the centre of gravity low. I concluded the vessel was much too small for the job it had to do. It was small because it was designed to fit existing (drydock) facilities.

These overruns placed enormous pressure on the project personnel and are difficult to cope with, ie. totally uncontrollable since the disciplines of mechanical, structural, electrical, instrumentation, hydraulic, drilling, diving, etc., etc., are so intermingled.

The total project cost for the Balmoral Field Development was £420,000,000. Today it produces 40,000 bbls/day and the rate of return on investment is +0.7%. An additional disadvantage (which is often overlooked) is that equipment inside the vessel cannot be removed to shore and must be maintained offshore. Operating cost is £40,000,000 per year which is extremely high.

I hope this answers your query.

Kind regards.



CRAIG LANG